

Assignment No. 7

Subqueries and Nested Queries

B.Sc. Data Science | Database Management Systems

**Q1) Create a MySQL database named CollegeDB and create two tables:
Course**

and StudentResult.

Course Table: CourseID (INT, PRIMARY KEY), CourseName (VARCHAR(50), NOT NULL), Fees (DECIMAL(10,2), NOT NULL)

StudentResult Table: StudentID (INT, PRIMARY KEY), StudentName (VARCHAR(50), NOT NULL), CourseID (INT, FOREIGN KEY), City (VARCHAR(30)), Marks (INT)

Insert at least 4 records into Course and 10 records into StudentResult.

CREATE DATABASE CollegeDB;

USE CollegeDB;

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50) NOT NULL,  
    Fees DECIMAL(10,2) NOT NULL  
);
```

```
CREATE TABLE StudentResult (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    CourseID INT,  
    City VARCHAR(30),  
    Marks INT,  
    FOREIGN KEY (CourseID) REFERENCES Course(CourseID)  
);
```

Q2) Write a MySQL query to display students whose marks are greater than the average marks of all students.

The output should contain:

Column

StudentName

Marks

ANS:

```
SELECT StudentName, Marks
```

```
FROM StudentResult
```

```
WHERE Marks > (
```

```
    SELECT AVG(Marks)
```

```
    FROM StudentResult
```

```
);
```

The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'collegedb' and its tables. The main panel displays the 'StudentResult' table with the following data:

StudentName	Marks
Sneha Joshi	91
Neha Shah	88
Meera Kulkarni	82
Divya More	93
Anita Kulkarni	84

Below the table, the console shows the executed query:

```
> SELECT StudentName, Marks FROM StudentResult WHERE Marks > ( SELECT AVG(Marks) FROM StudentResult );
```

Q3) Write a MySQL query to display all students enrolled in the BCA course using a subquery.

The output should contain:

Column

StudentID

StudentName

CourseID

Marks

ANS:

```
SELECT StudentID, StudentName, CourseID, Marks
FROM StudentResult
WHERE CourseID = (
    SELECT CourseID
    FROM Course
    WHERE CourseName = 'BCA'
);
```

The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'StudentResult' table is selected, and its structure is displayed. The table has four columns: StudentID, StudentName, CourseID, and Marks. Two rows are visible: (5, Karan Mehta, 102, 74) and (10, Divya More, 102, 93). Below the table, the 'Query results operations' section shows a console with two SQL queries. The first query is a subquery to find the average marks for each student. The second query is the main query to be executed, which filters the results to only include students from the 'BCA' course.

StudentID	StudentName	CourseID	Marks
5	Karan Mehta	102	74
10	Divya More	102	93

```
> SELECT StudentName, Marks FROM StudentResult WHERE Marks > ( SELECT AVG(Marks) FROM StudentResult );
> SELECT StudentID, StudentName, CourseID, Marks FROM StudentResult WHERE CourseID = ( SELECT CourseID FROM Course WHERE CourseName = 'BCA' );
```

Q4) Write a MySQL query to display each student with the overall average marks using a subquery in the SELECT clause.

The output should contain:

Column

StudentName

Marks

AverageMarks

ANS:

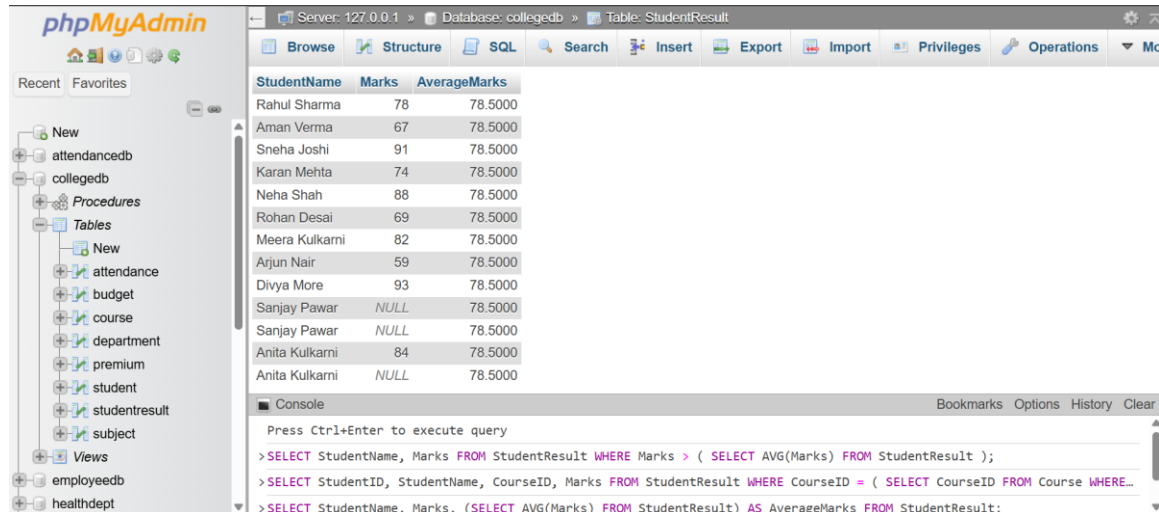
SELECT

StudentName,

Marks,

(SELECT AVG(Marks) FROM StudentResult) AS AverageMarks

FROM StudentResult;



StudentName	Marks	AverageMarks
Rahul Sharma	78	78.5000
Aman Verma	67	78.5000
Sneha Joshi	91	78.5000
Karan Mehta	74	78.5000
Neha Shah	88	78.5000
Rohan Desai	69	78.5000
Meera Kulkarni	82	78.5000
Arjun Nair	59	78.5000
Divya More	93	78.5000
Sanjay Pawar	NULL	78.5000
Sanjay Pawar	NULL	78.5000
Anita Kulkarni	84	78.5000
Anita Kulkarni	NULL	78.5000

```
> SELECT StudentName, Marks FROM StudentResult WHERE Marks > ( SELECT AVG(Marks) FROM StudentResult );  
> SELECT StudentID, StudentName, CourseID, Marks FROM StudentResult WHERE CourseID = ( SELECT CourseID FROM Course WHERE...  
> SELECT StudentName, Marks, (SELECT AVG(Marks) FROM StudentResult) AS AverageMarks FROM StudentResult;
```

Q5) Write a MySQL query using a correlated subquery to display students who scored more than their own course average.

The output should contain:

Column

StudentID

StudentName

CourseID

Marks

ANS:

SELECT

StudentID,

StudentName,

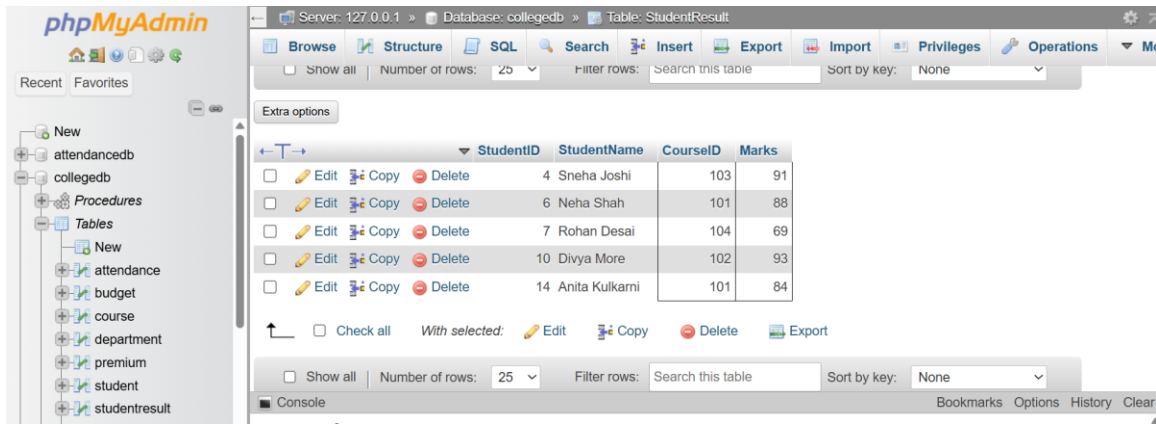
CourseID,

Marks

```

FROM StudentResult S1
WHERE Marks > (
    SELECT AVG(Marks)
    FROM StudentResult S2
    WHERE S2.CourseID = S1.CourseID
);

```



The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'StudentResult' table is selected, and its data is displayed in a table view. The table has four columns: StudentID, StudentName, CourseID, and Marks. There are five rows of data. The interface includes a sidebar with a tree view of the database structure, a top navigation bar with tabs like Browse, Structure, SQL, Search, Insert, Export, Import, Privileges, and Operations, and a bottom console area.

StudentID	StudentName	CourseID	Marks
4	Sneha Joshi	103	91
6	Neha Shah	101	88
7	Rohan Desai	104	69
10	Divya More	102	93
14	Anita Kulkarni	101	84

Q6) Write MySQL queries to display courses having students and courses

having no students.

Use the operators: EXISTS and NOT EXISTS

ANS:

1. Courses having students — EXISTS

```
SELECT CourseID, CourseName
```

```
FROM Course C
```

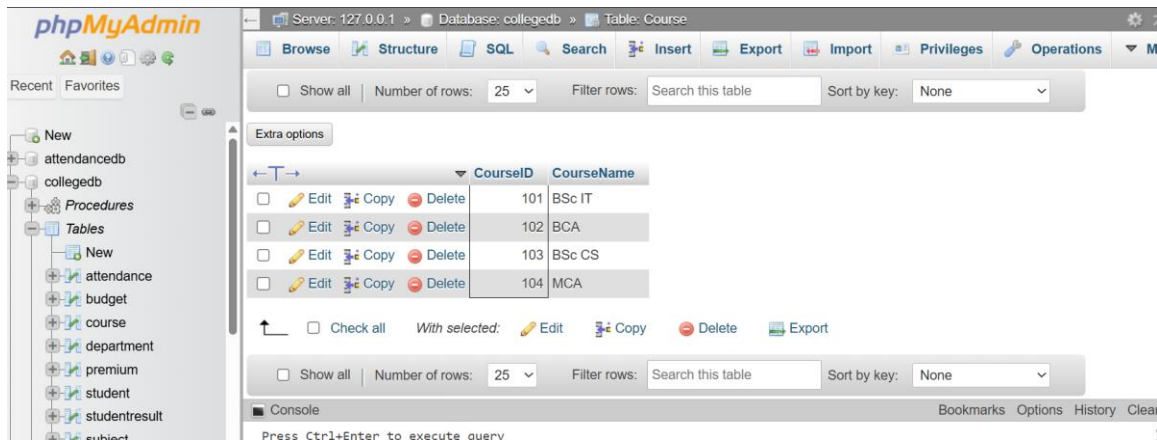
```
WHERE EXISTS (
```

```
    SELECT 1
```

```
    FROM StudentResult S
```

```
    WHERE S.CourseID = C.CourseID
```

```
);
```



2. Courses having no students — NOT EXISTS

SELECT CourseID, CourseName

FROM Course C

WHERE NOT EXISTS (

SELECT 1

FROM StudentResult S

WHERE S.CourseID = C.CourseID

);

